Research Article

ADVANCED PHOTONICS

Adaptive aberration-corrected quantitative phase microscopy via physics-informed deep learning

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Abstract. Quantitative phase microscopy (QPM) enables label-free imaging and precise characterization of transparent specimens by measuring phase delay. However, optical aberrations induce wavefront distortions that degrade phase reconstruction accuracy, resolution, and contrast. Existing strategies require diverse measurements or iterative optimization, limiting flexibility for real-time applications. We propose an adaptive aberration-corrected QPM system enabled by a physics-informed cycle-consistent network (PICNet) without prior calibration. By incorporating a learnable physical forward model to approximate the practical image formation and enforcing cycle consistency between object and measurement domains, PICNet can reconstruct the object phase from a single-shot measurement while simultaneously inferring complex aberrations that are difficult to characterize explicitly. Our approach achieves a 60% improvement in structural similarity compared with uncorrected results. Experiments demonstrate that PICNet enables rapid and high-fidelity phase retrieval across diverse biological samples with enhanced robustness to aberrations. This physically reliable and self-calibrating framework establishes a general paradigm for solving inverse problems across various computational imaging modalities.

Keywords: quantitative phase microscopy; deep learning; phase retrieval; biological imaging.

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1 Introduction

Optical microscopy has evolved into a powerful bioanalytical platform, enabling multidimensional structural and functional investigations of biological cells and tissues.[1] However, direct access to the biological details remains challenging because the inherent optical translucency and minimal light absorption of biological specimens lead to insufficient imaging contrast.[2] Fluorescence microscopy provides a potential solution to this limitation, allowing super-resolution imaging of targeted biological molecular structures through specific fluorescence labeling.[3] Despite its widespread use to enhance imaging contrast, exogenous labeling introduces phototoxicity and photobleaching, which disturb natural biological dynamics and prevent long-term longitudinal studies of live cells.[4] Quantitative phase microscopy (QPM), which can quantify phase delays introduced by the specimen, has proven to facilitate the visualization and characterization of viable cellular specimens in a label-free and non-destructive manner.[5] Recent advances in QPM have facilitated its widespread applications in diverse fields, such as biological imaging,[6] material metrology,[7] and medical diagnosis.[8] Reference-based interferometric QPM is commonly applied for phase reconstruction by analyzing fringe patterns produced through the interference between the object wave and the reference wave.[9] Nevertheless, the reliance on interferometric configurations[10,11] presents great challenges in achieving high spatial resolution and image quality, primarily due to the requirement for highly coherent light sources and complex optical setups.

Phase retrieval is designed to reconstruct the optical phase from intensity-only measurements, enabling reference-free and high-fidelity imaging.[12] In pursuit of improved spatial resolution, QPM with partially coherent illumination has been presented to reduce speckle noise, with examples including Fourier ptychographic microscopy[13–15] and transport of intensity equation.[16,17] The general phase-retrieval inverse problem can be formulated as a least-squares problem, which is solved by minimizing the loss between experimental measurements and

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physical model predictions. Alternating projection and convex optimization methods have been developed to provide effective solutions.[18] To mitigate the ill-posedness of phase retrieval, prior knowledge of the object field distribution is exploited as regularization terms, such as physical constraints[19,20] and sparsity priors.[21,22] Although such regularizers exhibit remarkable performance in improving reconstruction quality, the physical reliability of phase retrieval depends critically on the consistency between the idealized physical model and the practical imaging system. Ensuring consistency remains challenging due to the existence of deterministic and stochastic perturbations in physical imaging configurations.[23] The model mismatch problem in the imaging pipeline poses a critical challenge to robust experimental reconstructions.

Aberration is a ubiquitous perturbation in holographic microscopes, introducing wavefront distortions that affect the imaging process. Each type of aberration contributes a distinct perturbation to the object phase, thereby hindering accurate quantitative phase recovery and degrading both resolution and image contrast.[24] The aberration is induced by microscopic objectives,[25] off-axis recording,[26,27] and the refractive index structure of specimens.[28] Optical setup optimization[29,30] and numerical algorithms[31] are regarded as effective strategies for aberration compensation. Optical approaches typically rely on complex stacks of lens elements with precisely tailored parameters, enabling high-quality imaging with high temporal resolution at the expense of increased cost, weight, and complexity. By shifting the burden from hardware to post-acquisition algorithms, numerical correction has become a mainstream approach for suppressing aberrations such as spherical, astigmatism, and chromatic aberrations.[32,33] A common technique is pre-characterization of the optical system with subsequent computational removal via deconvolution, but this approach is limited by the trade-off between calibration complexity and computational cost.[34–36] Iterative phase retrieval with algorithmic self-calibration has been introduced to jointly estimate both the object phase and system aberrations, thereby avoiding complicated pre-calibration.[26,37–39] However, simultaneously decoupling phase and aberration contributions from intensity measurements requires diverse measurements[40] and regularization techniques with manually tuned parameters.[38] The numerical methods may fail under severe aberrations[41] and necessitate time-consuming inverse solvers, rendering them unsuitable for dynamic imaging. Therefore, automatic aberration correction from a single measurement is crucial for maintaining temporal resolution and accuracy of QPM.

Deep learning, featuring high-speed data processing and powerful statistical inference capabilities, has emerged as a compelling alternative to conventional approaches.[42] Rapid progress in deep learning algorithms has shown substantial potential to overcome the inherent limitations in traditional microscope systems,[43–45] achieving super-resolution imaging,[46] cross-modal image translation,[47,48] and virtual histological staining.[49] In particular, deep neural networks have proven effective for automatic aberration compensation in microscopy without explicit forward modeling.[50–52] The precise mapping from intensity to phase is learned by a large-scale paired dataset of aberration-distorted and aberration-free images.[53,54] Given the diversity and dynamic nature of biological specimens, acquiring sufficiently large paired datasets is both technically demanding and impractical. Such data-driven frameworks exhibit limited model generalization to out-of-distribution samples.[51] To eliminate the need for ground-truth datasets, untrained neural networks that incorporate physics-based models have been developed.[55–57] However, the convergence time of these methods typically ranges from minutes to hours.[58] As a result, the trade-off between computational efficiency and imaging accuracy remains a fundamental challenge for aberration-corrected QPM.

In this work, we propose an adaptive aberration-corrected QPM system enabled by a computationally efficient and physics-embedded deep learning framework, termed physics-informed cycle-consistent network (PICNet). PICNet leverages an aberration generator for blind aberration recovery and a phase generator for robust phase reconstruction from aberration-degraded measurements. Given that accurate phase reconstruction critically depends on precise wavefront propagation, the aberration generator is embedded into the physical forward model to enforce model consistency with the actual image formation, enabling the mapping from the measured data to aberration characteristics. Both generators are jointly optimized based on cycle consistency between the object and measurement domains, allowing PICNet to simultaneously recover object phase and pupil aberrations from a single-shot intensity pattern without prior calibration. PICNet exhibits improved fidelity compared with the uncorrected results. Unlike data-driven deep learning models that heavily rely on training datasets and lack physical guidance, PICNet exploits imaging physics to regularize network training, thus alleviating the overhead of dataset collection and enhancing the generalization ability across diverse imaging conditions. Experiments on quantitative phase targets and diverse biological specimens demonstrate the effectiveness of PICNet for aberration self-calibration. The major contribution of this work is to establish a general self-calibrating framework, addressing the long-standing challenge of physical model mismatch by fitting unknown parameters in the imaging process. The developed adaptive aberration-corrected microscopy system offers a promising path toward high-fidelity and real-time phase retrieval in biomedical imaging and optical metrology.

2 Results

2.1 Physics-informed Cycle-consistent Network

2.1.1 General framework

In microscopic imaging, the complex-amplitude signal is typically encoded into the intensity distribution at the sensor plane. Because electromagnetic waves oscillate beyond the speed of the detection devices, sensors can respond only to the intensity of the field, whereas the phase is lost during the measurement process. Thus, various phase retrieval algorithms have been developed to recover the object transmission function from intensity-only measurements.[21,59,60] The microscopic imaging process is schematically described in Fig. 1(a). The object, considered to be optically thin and transparent, is illuminated by the incident plane wave. The wavefront modulated by the complex-amplitude object is transferred to the pupil plane of the microscopic objective via a Fourier transform and subsequently relayed to the image plane by the tube lens through an inverse Fourier transform. The physical forward model of the image formation can be explicitly expressed as

$$\mathbf{I} = \mathcal{H}(\mathbf{o}, \mathbf{w}) = |\mathbf{F}^{-1}\mathbf{P}(\mathbf{w})\mathbf{F}\mathbf{o}|^2, \tag{1}$$

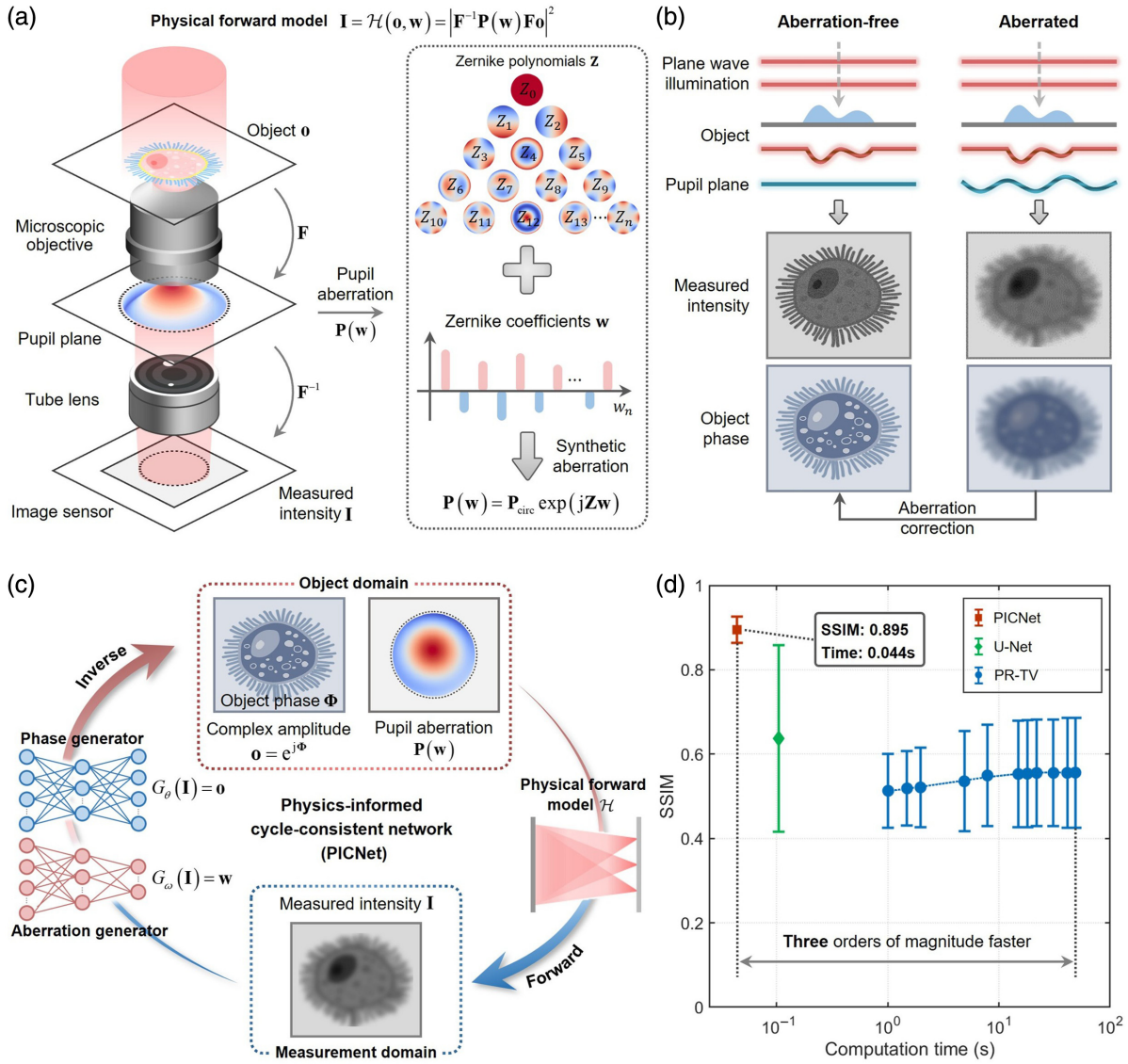


Fig. 1 Schematic illustration of the proposed PICNet framework. (a) Schematic diagram of physical forward model for the microscopic imaging system. (b) Comparisons of aberration-free and aberrated microscopy systems. Wavefront distortions introduced by optical aberrations lead to inaccurate phase reconstruction. (c) The working principle of PICNet. PICNet enables simultaneous inference of the sample phase and pupil aberrations from the given measurement. By enforcing cycle consistency, translation cycles including measurement–object–measurement and object–measurement–object are implemented to train the phase and aberration translators. (d) Comparative evaluation of algorithm runtime (measured on an Intel i7-13700KF CPU) and image fidelity. The markers represent mean values across 100 test samples, with error bars denoting the standard deviation.

where $\mathbf{o}$ represents the object transmission function, which is characterized as $\mathbf{o}=\exp(\mathrm{j}\mathbf{\Phi})$, and $\mathbf{\Phi}$ is the spatial distribution of object phase. $\mathbf{F}$ and $\mathbf{F}^{-1}$ denote the two-dimensional Fourier transform matrix and its inverse. $\mathcal{H}$ describes the physical forward operator. $\mathbf{I}$ is the measurement intensity. $\mathbf{P}$ is the pupil function. Aberrations in the imaging process can be modeled by incorporating a phase term into the coherent transfer function, with the synthetic pupil aberration represented as a linear combination of Zernike polynomials weighted by their respective coefficients,[24]

$$\mathbf{P}(\mathbf{w})=\mathbf{P}_{\text{circ}}\exp(\mathrm{j}\mathbf{Z}\mathbf{w}), \tag{2}$$

where $\mathbf{P}_{\text{circ}}$ is the ideal pupil function, which is a circular aperture with the cutoff frequency defined by $k\cdot\mathrm{NA}$, $k=2\pi/\lambda$, λ denotes the wavelength, and NA is the numerical aperture of the objective lens. $\mathbf{w}=(w_0,w_1,\ldots,w_n)^T$ is a vector containing the Zernike coefficients of each mode, and Zernike polynomials $\mathbf{Z}=(z_0,z_1,\ldots,z_n)$ consist of n orthogonal modes.

Ideally, the sample-induced optical phase delay under plane wave illumination can be accurately measured in the aberration-free imaging system, as illustrated in Fig. 1(b). However, lens-based microscopic systems inevitably suffer from intrinsic optical aberrations that distort the wavefront, leading to discrepancies between the reconstructed phase and the ground truth.

The effect of aberrations imposes fundamental constraints on imaging platform design. For example, conventional microscopes face an inherent trade-off between spatial resolution and field of view (FOV), primarily arising from aberrations induced by the objective lens. Although enlarging the lens's size can expand the FOV, this strategy inevitably introduces aberrations that degrade the spatial resolution and require complex optical corrections. The fundamental optomechanical limitations of conventional microscope system platforms make it particularly challenging to simultaneously achieve a wide FOV and high resolution in the design of the objective lens.[24] Physical limitations drive the development of algorithmic strategies for robust aberration correction, thereby ensuring faithful phase reconstruction.

In the aberrated QPM system, reconstruction of the object transmission function typically utilizes the penalized least squares (PLS) framework. The phase retrieval with total variation (PR-TV)[21] that combines a data-fidelity term $F(\mathbf{o})$ with a regularization term $R(\mathbf{o})$ takes the following form,

$$\hat{\mathbf{o}} = \arg\min_{\mathbf{o}} \{F(\mathbf{o}) + R(\mathbf{o})\}, \tag{3}$$

where $F(\mathbf{o}) = \frac{1}{2}\|\sqrt{\mathbf{I}} - |\mathbf{F}^{-1}\mathbf{P}(\mathbf{w})\mathbf{F}\mathbf{o}|\|_2^2$. In this optimization problem, prior characterization of system aberrations is essential to ensure that the forward model faithfully represents the physical imaging process. Without reliable aberration priors, reconstructions may fall into local minima, thereby degrading the reconstruction accuracy. Alternative approaches have been developed to jointly estimate the aberration distributions during the optimization process of phase retrieval algorithms, but multiple measurements with sufficient diversity are required to ensure a well-posed reconstruction.[37,61] Moreover, the optimization procedure within the PLS framework should be repeatedly performed whenever a new experimental measurement is acquired, which is unsuitable for inverse problems where fast reconstruction is necessary.

Inspired by recent advances in the optimal transport-driven CycleGAN (OT-CycleGAN) framework,[62–64] PICNet is proposed to address these challenges by leveraging deep learning to jointly predict the object phase and unknown pupil aberrations from a single-shot measurement. The proposed architecture incorporates deep neural networks into both the data-fidelity and regularization terms (see S1 in the Supplementary Material for analysis of the forward model with unknown physical parameters). As shown in Fig. 1(c), the aberration generator estimates unknown pupil aberrations directly from the measurement, thereby addressing inverse problems involving unknown parameters in the forward model. In addition, the proposed method addresses the inverse problem by incorporating a phase generator to recover the object phase from the measurement. Consequently, adaptive aberration-corrected phase imaging provides a more faithful recovery of the object phase distribution, whereas methods without correction are inevitably compromised by aberration-induced distortions. Specifically, the proposed model is trained with two translation cycles between object and measurement domains. In one cycle, a measurement $\mathbf{I}$ is translated through the aberration generator G_ω and the phase generator G_θ to predict both the object transmission function $\mathbf{o}$ and the pupil aberrations $\mathbf{P}(\mathbf{w})$, which are then passed through the physical forward model $\mathcal{H}$ to return to the measurement domain. In another cycle, the object $\mathbf{o}$ with a given pupil aberration $\mathbf{P}(\mathbf{w})$ is propagated through a parameterized forward model $\mathcal{H}$ to synthesize a measurement $\mathbf{I}$, which is then translated back to the object domain via the two generators G_θ and G_ω. The object phase $\mathbf{\Phi}$ is the argument of $\mathbf{o}$. This framework enforces cycle consistency to guarantee that round-trip mappings accurately recover the original input, thereby regularizing the generators and enabling faithful phase recovery despite unknown aberrations.

Figure 1(d) compares the structural similarity index measure (SSIM) and computation runtime of the proposed PICNet against PR-TV and U-Net methods on 100 test samples. The PR-TV method that tackles the phase retrieval without considering aberration correction is employed, using gradient-domain sparsity as the regularization prior. The PR-TV method exhibits limited improvement in reconstruction quality with increasing iterations and ultimately converges to an average SSIM of 0.5561 at a computational cost of 48 s. As a data-driven deep learning model, the U-Net attains an average SSIM of 0.6374 with a processing time of 0.1 s through supervised learning of the inverse mapping between input and corresponding label data. However, its substantially larger standard deviation reveals crucial sensitivity to input diversity. The proposed PICNet enables rapid phase reconstruction in only 0.04 s while maintaining superior fidelity with an average SSIM of 0.8947. Compared with PR-TV with iterative optimization, PICNet achieves a speedup of nearly three orders of magnitude. In terms of quality, PICNet improves the structural similarity by 60% compared with PR-TV without aberration correction and 40% relative to U-Net. Moreover, PICNet exhibits significantly lower standard deviation than U-Net, indicating superior generalization performance across diverse inputs. Consequently, PICNet demonstrates its potential for achieving high-speed and high-fidelity phase reconstruction with robust generalization.

2.1.2 Training procedure

During training, PICNet learns a mapping between measurement and object domains with unpaired intensity and object phase data. Without explicit paired supervision, relying on a single translation cycle may lead to poor convergence and mode collapse. Therefore, a dual-cycle translation framework is introduced, enabling robust phase reconstruction and precise aberration correction in unpaired training settings. The proposed framework jointly optimizes the phase and aberration generators (for details about the network architecture, see Sec. 4) by enforcing cycle consistency, which mitigates training imbalance in which one generator dominates the optimization while the other collapses. A physics-based forward model further restricts the solution space by enforcing physical validity of the generated outputs, thereby enhancing training stability and enabling effective decoupling between object phase and pupil aberrations. To further constrain the space of possible mappings, an adversarial discriminator is introduced to distinguish between the ground truth phase and the retrieved one generated by the phase generator (see S2 in the Supplementary Material for detailed discussions on the contribution of the discriminator). By enforcing adversarial losses, the discriminator regularizes the phase generator and ensures statistical consistency with the true phase distribution. This strategy promotes balanced convergence of both generators, improves generalization across varying imaging conditions, and thereby enables a more reliable inverse mapping.

Figure 2(a) presents the training process of the translation cycle from the measurement domain to the object domain and back to the measurement domain. Given the input measured intensity $\mathbf{I}$, the generators G_θ and G_ω estimate the object

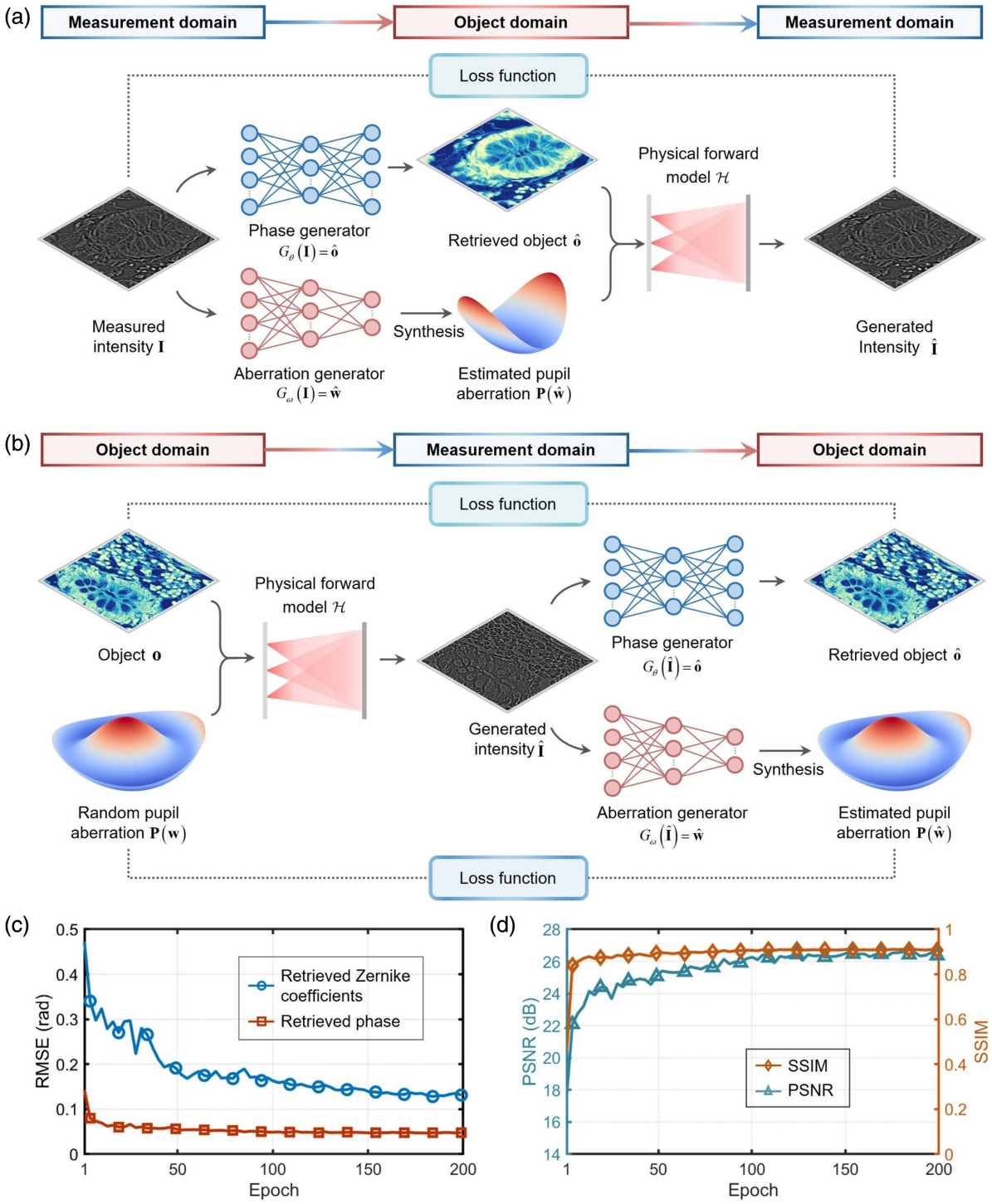


Fig. 2 Training procedure of PICNet and performance evaluation. (a) The training process of the translation cycle from the measurement domain to the object domain and back to the measurement domain. (b) The training process of the translation cycle from the object domain to the measurement domain and back to the object domain. (c) Evolution of the RMSE values for the retrieved phase and Zernike coefficients versus epochs on 100 test images. (d) Optimization curves showing PSNR and SSIM of the retrieved phase versus epochs on 100 test images.

transmission function and pupil aberrations, which are subsequently utilized to simulate the output intensity distribution $\hat{\mathbf{I}}$ through a physical forward model $\mathcal{H}$. Cycle-consistency loss between the measured $\mathbf{I}$ and generated intensities $\hat{\mathbf{I}}$ is minimized to optimize the parameters of two networks. Conversely, the training process of the translation cycle in Fig. 2(b) is from the object domain to the measurement domain and returns to the object domain. By combining the given object $\mathbf{o}$ with randomly generated

pupil aberrations $\mathbf{P}(\mathbf{w})$, the intensity pattern can be synthesized based on the forward operator $\mathcal{H}$. This synthesized intensity pattern is then employed to predict both the object $\hat{\mathbf{o}}$ and the pupil aberration profile $\mathbf{P}(\hat{\mathbf{w}})$ through the networks G_θ and G_ω. The optimization of network parameters is guided by two separate cycle-consistency losses, imposed between the predicted $\hat{\mathbf{o}}$ and ground-truth object $\mathbf{o}$ and between the predicted $\mathbf{P}(\hat{\mathbf{w}})$ and ground-truth pupil aberrations $\mathbf{P}(\mathbf{w})$, respectively.

To quantitatively assess the performance of the proposed network during training, a set of 100 test images is employed for evaluation at each training epoch. As seen in Fig. 2(c), the root mean squared error (RMSE) between the retrieved and ground-truth phase profiles, as well as between the predicted and true Zernike coefficients, gradually decreases during training and converges to 0.04 and 0.12, respectively. This trend reflects stable optimization and effective learning of the underlying mappings from intensity measurements to aberration representations. The fidelity of the retrieved phase is evaluated using the peak signal-to-noise ratio (PSNR) and the SSIM, as presented in Fig. 2(d). Both PSNR and SSIM steadily increase during training and converge to average values of 26.55 dB and 0.9097, respectively. Results demonstrate that PICNet reconstructs the quantitative phase and infers the corresponding pupil aberrations in a robust and generalizable manner.

2.2 Quantitative Phase Microscopy Using PICNet

2.2.1 Performance evaluation in simulation

The PICNet model is trained on a dataset generated using Zernike polynomials, with coefficients randomly sampled from a uniform distribution under the small-defocus condition (see Sec. 4 for more details). The model is evaluated on intensity patterns synthesized using Zernike coefficients drawn from the same distribution as the training set and from an extended range. As shown in Fig. 3(a), the proposed method accurately reconstructs the phase from intensity patterns affected by aberrations both within and beyond the training distribution, corresponding to measurement A and measurement B, respectively. Compared with the ground truth, the retrieved phase for measurement A achieves a PSNR of 25.96 dB and an SSIM of 0.91, while that for measurement B reaches 24.41 dB and an SSIM of 0.85, demonstrating high-fidelity reconstruction.

Further analysis in Fig. 3(b) illustrates that the retrieved phase profiles are in close agreement with the ground truth. In addition to phase reconstruction, PICNet simultaneously predicts Zernike coefficients for adaptive aberration correction, whereas conventional methods typically rely on prior aberration characterization followed by computational compensation.[65] The performance of aberration prediction is evaluated in Figs. 3(c) and 3(d), where the reconstructed pupil aberrations are in agreement with the ground truth. Different methods on test datasets with pupil aberrations beyond the training distribution are compared and analyzed (see S3 in the Supplementary Material). The proposed method facilitates self-calibration and robust phase recovery from intensity measurements perturbed by aberrations.

The reconstruction performance of the proposed PICNet model, in terms of fidelity and generalization capability, is quantitatively compared with two representative methods, PR-TV and U-Net. PR-TV is an iterative algorithm for phase retrieval relying on a proximal gradient descent scheme. Specifically,

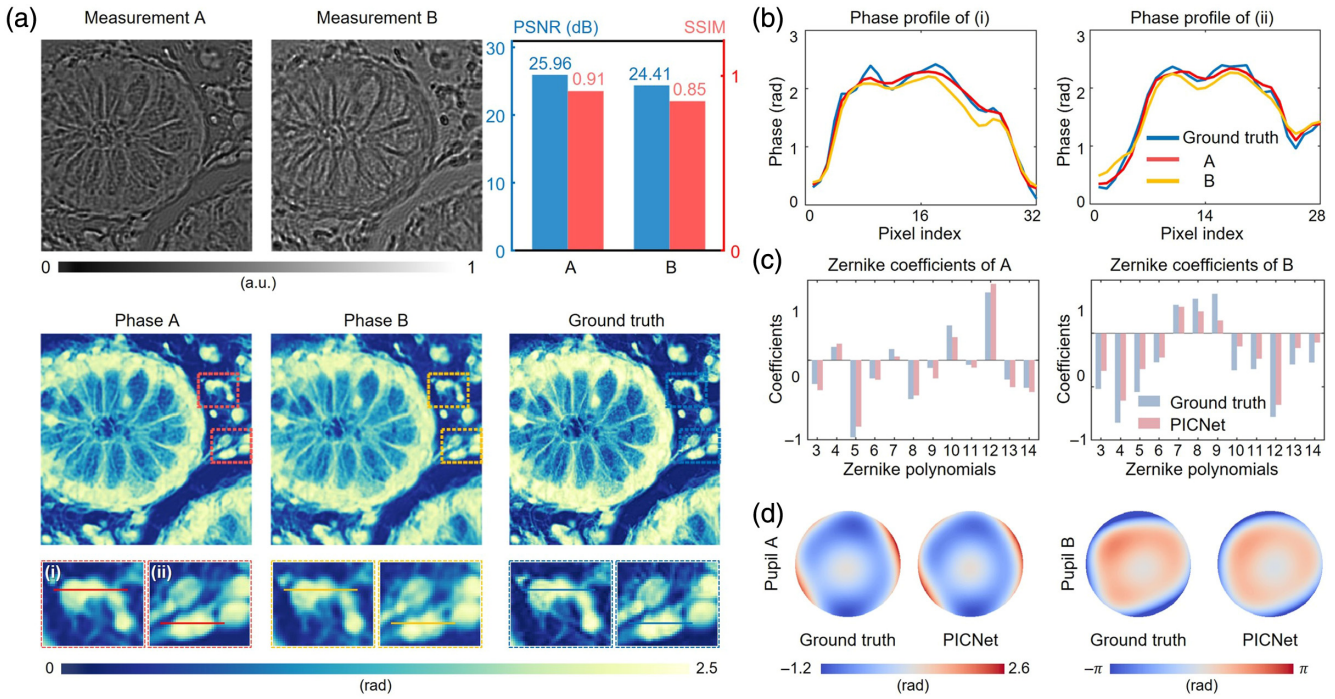


Fig. 3 Simulation verification of the PICNet to simultaneously reconstruct the phase and pupil aberrations. (a) Measurements with various aberrations, retrieved phase maps, ground truth phase maps, and corresponding PSNR and SSIM metrics. (b) Reconstructed and ground truth phase profiles along the solid lines in the magnified view of (i) and (ii) in panel (a). (c) Predicted and ground truth Zernike coefficients. (d) Reconstructed and ground truth pupil aberrations. A denotes aberrations within the training data distribution range, whereas B denotes those beyond this range.

such a method is implemented by minimizing the data-fidelity term via gradient descent and updating the regularization term by its proximity operator. U-Net represents a data-driven deep learning framework, trained end-to-end using paired source-target domain datasets without introducing an explicit physical model. U-Net is used for comparison to investigate the impact of excluding the physical model on phase reconstruction accuracy. PICNet leverages a physics-informed deep learning framework and is trained via a dual-cycle translation strategy between the source and target domains with unpaired datasets. Both U-Net and PICNet models are trained on normal tissue datasets with Zernike coefficients sampled from an identical distribution under the large-defocus condition (see Sec. 4 for more details).

The reconstruction quality of various methods is evaluated in Fig. 4(a), with test datasets generated by normal tissues.

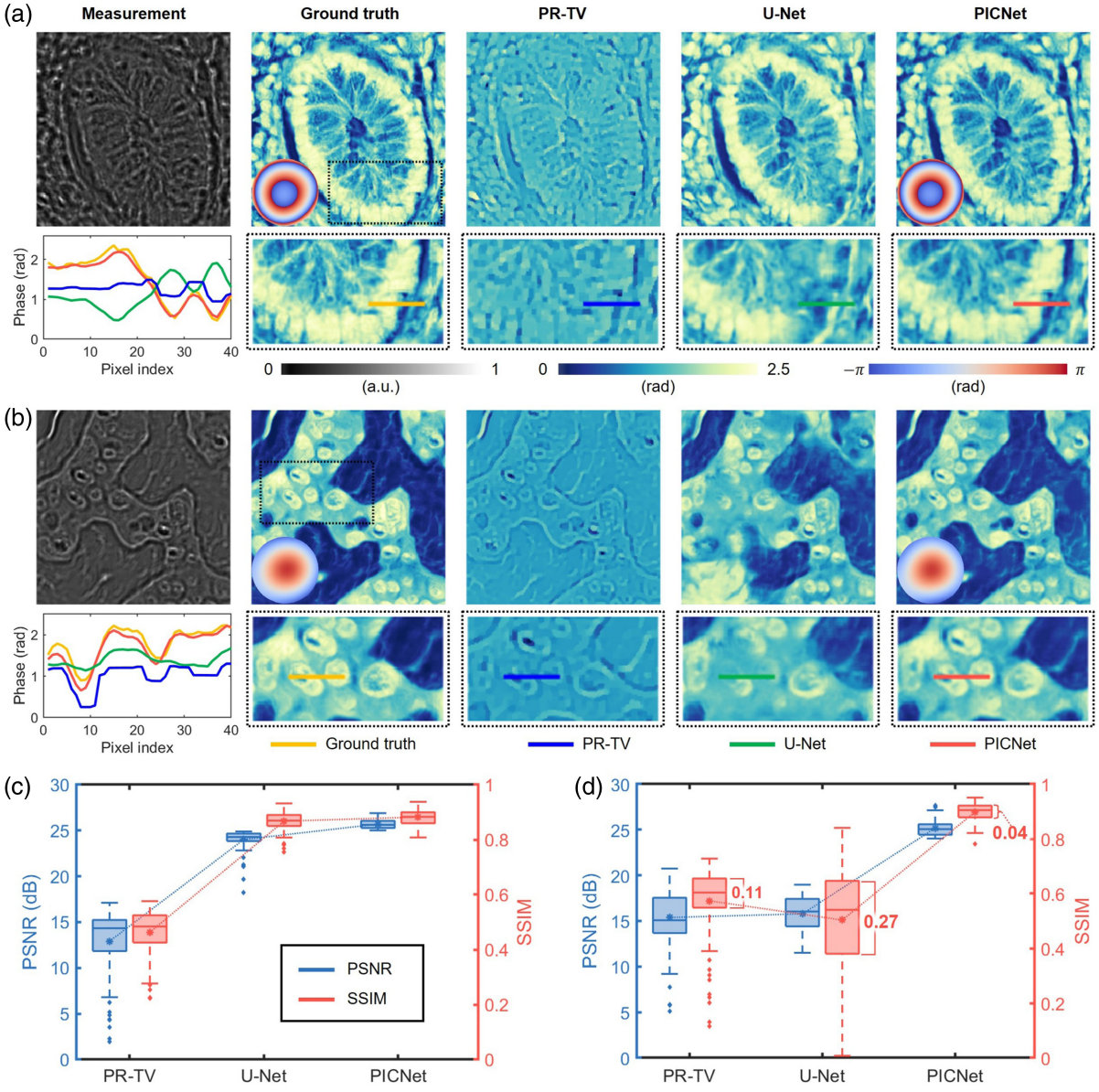


Fig. 4 Reconstructions obtained using different methods. Measured intensity, ground truth, reconstructed phase images obtained using various methods, and cross-sectional phase profiles along the solid lines in the magnified view for (a) the normal colon mucosa tissue and (b) the breast cancer tissue. The lower right corner of the images indicates the pupil aberrations obtained by PICNet and the corresponding ground truth. Boxplots of the PSNR and SSIM values of the reconstructed phase images for (c) the normal colon mucosa tissue and (d) the breast cancer tissue, evaluated over 100 test samples using PR-TV, U-Net, and PICNet. On each box, the central line denotes the median, and the lower and upper edges of the box represent the 25th and 75th percentiles, respectively. Whiskers extend to the most extreme data points not considered outliers, and the outliers are plotted with the diamond marker symbol. The mean values are marked with asterisks and connected by dotted lines.

The PR-TV method without correcting aberrations fails to recover the sample phase from intensity measurements perturbed by aberrations, resulting from the inaccurate simulation of the forward model in the optimization of the inverse problem. Quantitative comparisons with the ground truth indicate that phase reconstructions obtained via U-Net suffer from artifacts. By contrast, the PICNet model predicts the pupil aberrations with a low RMSE of 0.137 rad and achieves improved reconstruction fidelity with a PSNR of 26.74 dB.

To further verify the generalization capability, an independent dataset based on breast cancer tissues[66] is employed to compare the reconstruction quality of different methods, as shown in Fig. 4(b). Notably, this dataset is distinct from the training set that comprises normal tissues. As U-Net solely relies on given datasets for training, its learned mapping from measured intensity to sample phase is inherently limited to the characteristics present in the training set. When applied to test data with different morphological and structural features, the reconstruction fidelity deteriorates substantially, exhibiting significant discrepancies in cellular structure compared with the ground truth. By contrast, PICNet demonstrates more robust phase reconstruction, producing predictions that closely match the ground truth in both phase accuracy and structural fidelity, with a high PSNR of 24.61 dB. The pupil aberration retrieved by PICNet shows good agreement with the ground truth, achieving an RMSE of 0.148 rad.

The PSNR and SSIM metrics of the phase map recovered by the PR-TV, U-Net, and PICNet methods across 100 test samples are presented in Figs. 4(c) and 4(d). On the normal colon mucosa tissue dataset, PICNet achieves a mean PSNR of 1.68 dB higher and a mean SSIM of 0.015 higher than those of U-Net, indicating comparable reconstruction fidelity between the two methods. However, when tested on the breast cancer tissue dataset, PICNet significantly outperforms U-Net, achieving a 9.4 dB improvement in mean PSNR and a 0.395 increase in mean SSIM. Moreover, the interquartile range of SSIM for U-Net reaches 0.27, significantly larger than the 0.04 observed for PICNet, illustrating larger dispersion and lower generalization of U-Net. It is evident that PICNet has superior generalization capability, enabling robust phase retrieval across samples with diverse structural characteristics.

2.2.2 Experimental setup and quantitative characterization

To experimentally validate the effectiveness of the proposed PICNet framework, an adaptive aberration-corrected QPM system for imaging transparent phase targets and biological specimens is constructed. The experimental setup follows the diagram in Fig. 5(a). To eliminate speckle in laser illumination, a light-emitting diode (LED) with a nominal wavelength of 625 nm is used as the light source for illumination. The LED illumination is positioned sufficiently far away to ensure plane wave approximation at the object plane. To enhance spatial coherence, a pinhole (Thorlabs, P500K) with a diameter of 500 μm is fixed after the LED. A bandpass filter (Thorlabs, FBH630-10) with a 10-nm bandwidth is placed after the pinhole to improve temporal coherence.[67] The transmitted light from the sample is collected by a microscopic objective (Mitutoyo, 10×, NA = 0.28) and then relayed by an Olympus tube lens with a focal length of 180 mm. The sample is mounted on a translation stage. Defocus distances are precisely controlled

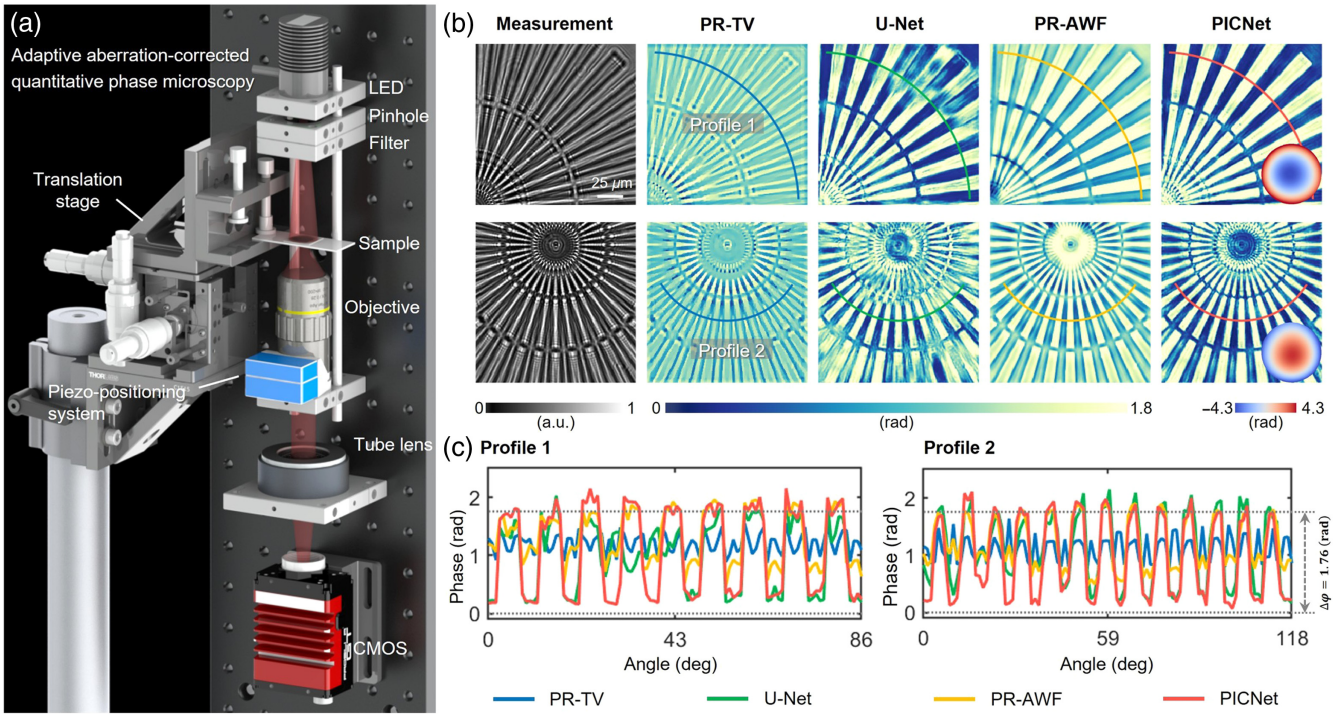


Fig. 5 Experimental setup and characterization of a quantitative phase target. (a) Optical configuration of the adaptive aberration-corrected QPM system. (b) Experimental comparison of different phase retrieval methods at defocus distances of −30 and 25 μm, as well as the retrieved pupil aberration obtained by PICNet. (c) Cross-sectional phase profiles along the marked lines in panel (b), where the gray-dashed lines indicate the theoretical phase value.

using a piezoelectric positioning stage with a travel range of 100 μm, which adjusts the axial position of the objective. The final intensity distributions are recorded by a CMOS sensor (QHY174), featuring a 5.86 μm pixel size and a resolution of 1920×1200 pixels. The system achieves a spatial resolution of 1.36 μm, with an FOV of $\sim$1.25 mm $\times$ 0.78 mm. The captured intensity is then digitally processed for phase reconstruction with adaptive aberration correction. In the quantitative phase imaging system, measurement noise is indeed inevitable. Hence, Gaussian noise is added to the intensity images of the dataset to enhance the noise robustness of PICNet (see S4 in the Supplementary Material).

The performance of the designed imaging system is evaluated using a quantitative phase target (QPT, Benchmark Technologies). In addition to PR-TV and U-Net, phase retrieval using the accelerated Wirtinger flow algorithm (PR-AWF) is included as a reference,[68] which employs six measurements at different known defocus distances. Figure 5(b) shows the raw intensity recorded at defocus distances of 30 and 25 μm and the corresponding phase reconstructions obtained by different methods for comparison. Both U-Net and PICNet are trained on a dataset of normal tissues, with Zernike coefficients defined under the large-defocus condition. PR-TV fails to accurately reconstruct the phase structure and exhibits prominent artifacts. This can be attributed to its strong reliance on the data-fidelity term derived from a physical model, which cannot fully account for system-specific aberrations. Such intrinsic aberrations in practical optical systems introduce a mismatch between the simulated forward model and the actual imaging process, thereby rendering the phase retrieval physically unreliable. As for the U-Net with limited generalization ability, the feature differences between the training data and the QPT make it fail to reconstruct the phase distributions of QPT. Although PR-AWF with multiple known defocus distances as priors provides improved accuracy in phase reconstruction, the correction is confined to defocus, leaving the results susceptible to other aberrations. Notably, PICNet outperforms PR-TV, U-Net, and PR-AWF, recovering smooth staircase phase profiles with sharp edges. The measured resolution achieved by PICNet is 1.38 μm, closely approaching the theoretical resolution (see S5 in the Supplementary Material for more details). According to the recovered pupil aberrations, the quadratic defocus aberration is the dominant component contributing to the system's phase distortion.

Given that the object possesses a uniform refractive index distribution and the fabricated features reside solely on the surface of the substrate, the height profiles can be directly calculated and characterized by

$$\Delta\Phi = \frac{2\pi}{\lambda}(r - r_o)h, \tag{4}$$

where $\Delta\Phi$ denotes the phase delay, r is the refractive index of the medium, r_o is the refractive index of the surroundings, and h represents the height of the structure. Hence, the surface height of the microstructures can be directly inferred from the reconstructed phase map. As shown in Fig. 5(c), phase profiles retrieved by PICNet closely approximate the ground-truth phase delay of 1.76 rad that is calculated based on the known structure height of 0.336 μm and material refractive index of 1.52. By comparison, phase profiles measured with PR-TV, U-Net, and PR-AWF exhibit large deviations from the ground-truth values. These findings demonstrate the quantitative phase imaging ability of the designed imaging system.

2.2.3 Experimental validation for applications in biological imaging

To assess the generalization ability and robustness of PICNet in biological phase imaging, we evaluate its performance across a range of representative specimens. Figures 6(a)–6(d) present raw intensity measurements, reconstructions obtained using various phase retrieval algorithms, and the corresponding H&E-stained sections. The H&E-stained images are included as references, revealing the underlying biological structure. Reconstructions obtained from PR-TV, U-Net, and PR-AWF suffer from noise and artifacts, whereas PICNet produces phase images with improved structural fidelity, showing close agreement with the in-focus brightfield microscopy images of the H&E-stained sections. Phase retrieval with PICNet enables clear visualization of morphological structures in biological tissues. Simultaneously, unknown pupil aberrations are recovered by the aberration generator of PICNet, facilitating blind compensation of aberration-induced distortions. The quantitative phase reconstructions provide precise measurements of optical path length variations across the sample, enabling high-contrast imaging of weakly scattering biological samples. High-quality reconstructions of biological specimens with diverse structural features, including small intestine samples, gastric wall tissues, and porcine hepatocytes, highlight the strong generalization capability of PICNet. These experimental observations reveal that PICNet offers a powerful solution for high-fidelity phase imaging in a wide range of samples, holding great potential for applications in digital pathology and live-cell imaging where adaptive aberration correction is essential.

To further demonstrate the aberration self-calibration performance of the proposed PICNet framework, different aberration levels are imposed on the microscopy system by systematically defocusing the sample at varying distances. The piezoelectric positioning stage is employed to scan along the optical axis in steps of 5 μm. The intensity images are captured at each axial position to evaluate phase retrieval under different defocus conditions. The corresponding reconstructed phase and estimated pupil aberrations are presented in Fig. 7(a). For comparison, different phase retrieval algorithms including PR-TV, U-Net, and PICNet are implemented.

Among these three approaches, the phase images recovered by PICNet maintain a higher contrast while effectively suppressing the artifacts under different defocus conditions. In particular, PICNet successfully compensates for aberrations with a maximal phase deviation approaching 2.7π, corresponding to a defocus of 30 μm, highlighting its robustness against severe optical aberrations. PR-TV exhibits rapid degradation in reconstruction quality as the defocus increases, highlighting its vulnerability to severe aberration perturbations. This degradation is attributed to the method's reliance on iterative optimization without prior aberration knowledge, resulting in convergence to local minima. Although in principle axial positions can be estimated using autofocusing algorithms[69–71] to correct for defocus, PR-TV remains highly susceptible to noise and additional aberrations arising from mechanical misalignments, refractive index inhomogeneities, and imperfect optical components. Meanwhile, phase images recovered by U-Net suffer from artifacts and noises, leading to blurred and indistinct cellular structures. This performance limitation results from the absence of explicit physical constraints in purely data-driven models.

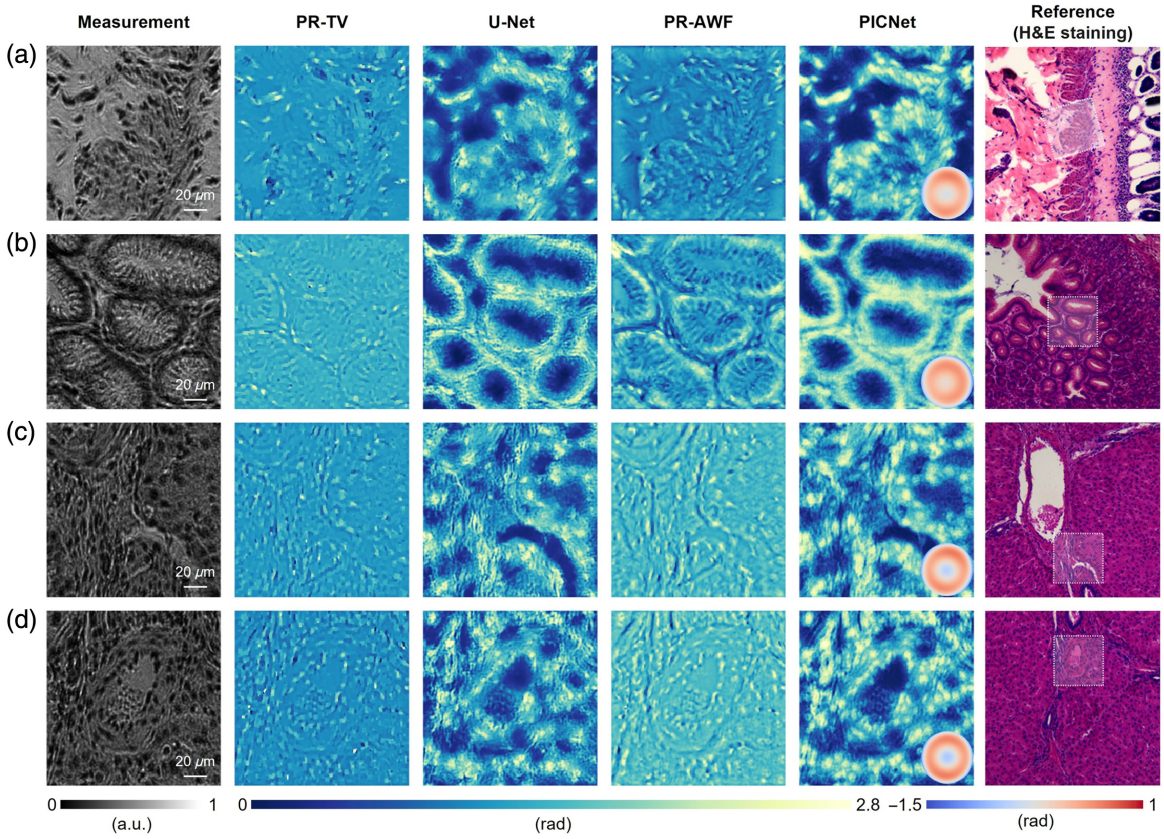


Fig. 6 Adaptive aberration-corrected QPM for different types of biological specimens. Measurements, retrieved phase profiles obtained using different methods, and the corresponding H&E-stained images of (a) small intestine, (b) gastric wall, and (c), (d) porcine hepatocyte samples. Shown in the lower-right corner of each PICNet-retrieved phase image are the estimated pupil aberrations. The retrieved phase corresponds to the area outlined by the dashed white rectangle in the H&E-stained image.

Under large defocus conditions, the phase aberration in the pupil plane is dominated by the Zernike defocus mode, characterized by a quadratic phase profile. As expected, the aberrations recovered by PICNet accurately exhibit this quadratic form, with increasing phase curvature corresponding to larger axial displacements, which is shown in Fig. 7(b). The estimated Zernike coefficients of the defocus mode through PICNet versus the experimentally measured defocus distances are exhibited in Fig. 7(c). A linear regression analysis between the estimated Zernike coefficients of the defocus mode and the defocus distances reveals a strong correlation ($R^2 = 0.9896$), confirming the quantitative reliability of the aberration estimation by PICNet. Our findings demonstrate that PICNet enables robust, high-fidelity, and artifact-free phase reconstruction under large defocus without requiring prior knowledge of system aberrations. The adaptive calibration capability of PICNet suggests that it is a promising candidate for blind correction of complex and dynamically varying aberrations.

3 Discussion and Conclusion

From an algorithmic perspective, PICNet provides an effective framework for solving inverse problems in computational imaging, where forward model parameters are unknown or difficult to characterize. By embedding a learnable physical forward model into the training pipeline, PICNet constrains the solution space and guides the network toward physically consistent outputs. This strategy reduces the mismatch between idealized models and practical imaging conditions, offering a robust and generalized framework to handle perturbations in imaging systems.[72,73] For example, the mapping from intensity to pupil aberrations can be inferred without explicit pre-calibration in aberration-affected microscopy. PICNet with blind aberration estimation supports high-fidelity and high-temporal-resolution quantitative phase imaging in dynamic scenarios (see S6 in the Supplementary Material for more details about the comparison of different methods).

From an application perspective, PICNet facilitates real-time and high-fidelity observation in biomedical imaging[74,75] and optical metrology,[76,77] thereby contributing to the development of intelligent computational microscopy systems.[78] Although this work primarily focuses on adaptive aberration correction in quantitative phase imaging, the proposed framework can also be regarded as a general paradigm for inverse problem-solving across other computational imaging modalities, such as fluorescence microscopy,[79,80] computed tomography,[81] computational ghost imaging,[82] and astronomical imaging.[83] Despite their differences, these imaging systems share a common characteristic: they are governed by a well-defined physical forward model, which can be modeled through principles of diffraction,

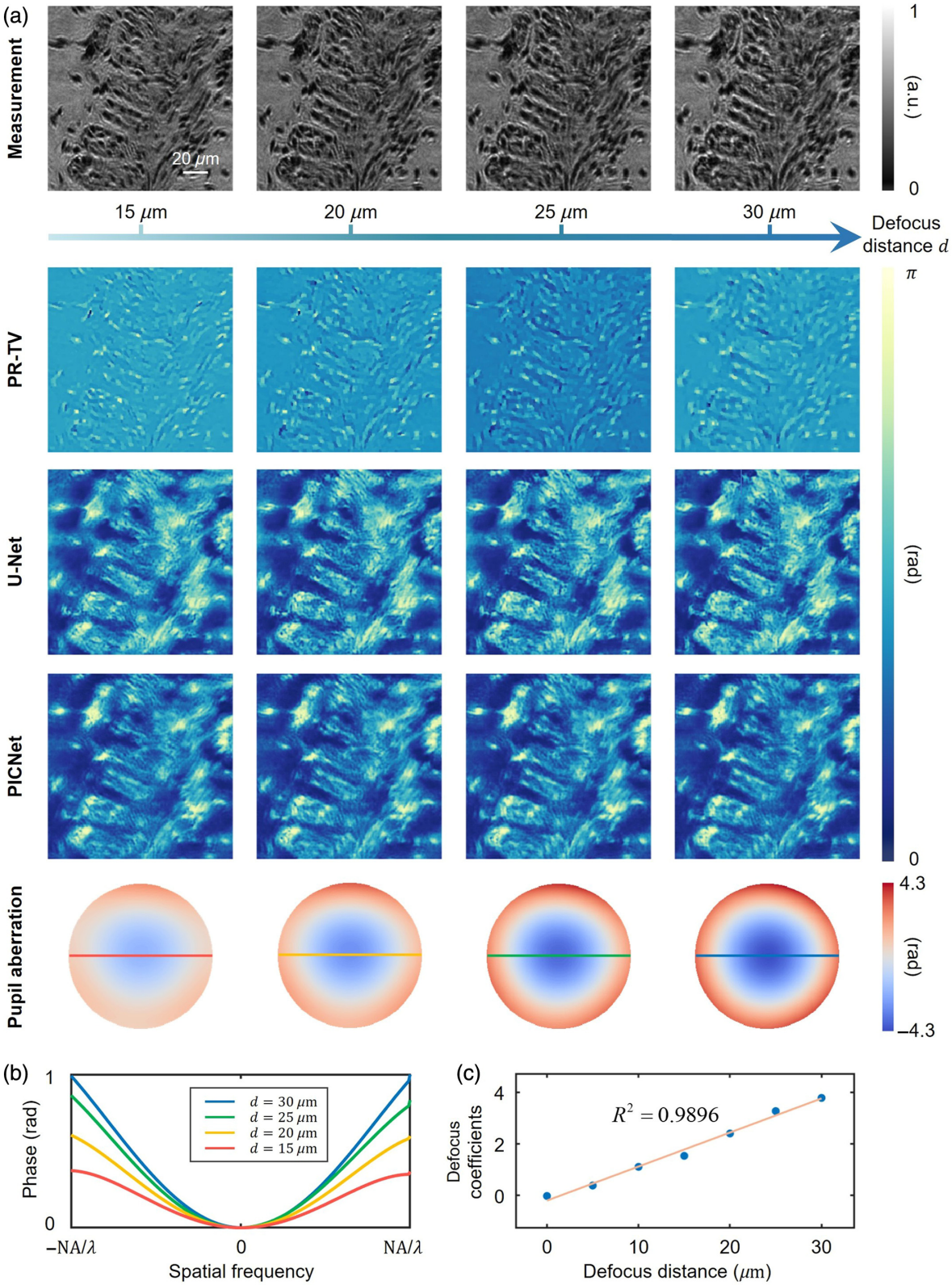


Fig. 7 Experimental validation of adaptive aberration correction. (a) Measurements, retrieved phase images by different methods, and recovered pupil aberrations at various defocus distances. (b) Normalized cross-sections of the pupil aberrations extracted along the solid lines in panel (a). (c) Estimated defocus parameters compared against experimentally measured distance values ($R^2 = 0.9896$). d denotes the defocus distance.

interference, scattering, and light–matter interaction, whereas unknown physical parameters can be adaptively learned by neural networks without prior characterization. We envision that the general principle underlying PICNet holds great potential for advancing computational imaging, enabling solutions beyond the scope of conventional approaches.

In conclusion, we have developed a QPM system with adaptive aberration correction by using PICNet, a physics-informed cycle-consistent deep learning framework. By integrating an aberration generator into a differentiable physical forward model and coupling it with a phase reconstruction network, PICNet enables blind retrieval of both object phase and pupil aberrations directly from a single-shot intensity measurement. The physical forward model is able to learn unknown aberration parameters, ensuring consistency between the image formation process and its computational simulation without requiring explicit priors. The two networks are jointly optimized under a cycle-consistency constraint between the object and measurement domains. Numerical simulations and experimental results validate the algorithmic performance, demonstrating superior quality, faster computation, better generalization, and improved robustness against aberrations compared with conventional phase retrieval and data-driven deep learning methods. Beyond QPM, PICNet offers a generalizable solution to inverse problems in computational imaging, where system imperfections and model uncertainties often degrade performance. The proposed paradigm opens new avenues for self-calibrating, real-time, high-performance phase imaging in biomedical and industrial applications.

4 Appendix: Materials and Methods

4.1 Data Preparation and Training Details

Most existing data-driven deep learning approaches for phase retrieval rely on paired datasets consisting of intensity measurements and their corresponding phase ground truth.[84] By contrast, PICNet is designed to learn the statistical mapping between the measurement domain and the object domain in an unpaired manner, thereby eliminating the requirement for simultaneously acquired intensity–phase image pairs. This unpaired learning paradigm substantially broadens the applicability of phase retrieval networks, particularly in scenarios where collecting paired training data is challenging or infeasible. To train and validate the proposed framework, we employed the NCT-CRC-HE-100K histopathology dataset.[85] After processing, 24,000 measured intensity images and 5098 object phase images at a spatial resolution of 224×224 pixels are obtained. Notably, the dataset encompasses diverse biological tissue types, including smooth muscle, normal colon mucosa, and lymphocytes. PICNet demonstrates robust generalization in the retrieval phase for both quantitative phase targets and biological samples. The test dataset consists of images that are not included during training, ensuring an objective evaluation of the model's performance.

As described in Eq. (2), pupil aberrations are modeled as a weighted linear combination of Zernike polynomials. We generate synthetic pupil aberrations by sampling random Zernike coefficients, with the associated coefficients w_n randomly sampled within a predefined upper bound V_n:

$$|w_n| \le V_n, \quad n = 1, 2, \ldots, N, \tag{5}$$

where n denotes the Zernike polynomial index according to the OSA convention,[86] covering piston ($n = 0$), tilt ($n = 1, 2$), defocus ($n = 4$), astigmatism ($n = 3, 5$), coma ($n = 7, 8$), trefoil ($n = 6, 9$), primary spherical ($n = 12$), tetrafoil ($n = 10$, 14), and secondary astigmatism ($n = 11, 13$), which are described in Fig. 1(a). N represents the maximum Zernike index, which is determined by the chosen Zernike order:

$$N = \begin{cases} 9, & \text{for Zernike order of 3} \\ 14, & \text{for Zernike order of 4} \end{cases}. \tag{6}$$

To construct the training dataset, two aberration scenarios are considered, referred to as the small-defocus and large-defocus conditions. In the small-defocus case, aberrations are included up to the 4th Zernike order ($N = 14$). The piston and tilt components ($n = 0, 1, 2$) are excluded, and the remaining coefficients are initially constrained to an upper bound of 0.5 rad. To better approximate practical imaging conditions, an additional 0.5 rad for astigmatism ($n = 3, 5$) and an additional 1 rad for defocus ($n = 4$) and primary spherical ($n = 12$) are added, as these terms typically induce more severe distortions in the QPM system. Accordingly, the upper bound V_n was defined as

$$V_n = \begin{cases} 0, & \text{for } n = 0, 1, 2 \\ 1, & \text{for } n = 3, 5 \\ 1.5, & \text{for } n = 4, 12 \\ 0.5, & \text{otherwise for } n \le N \end{cases}. \tag{7}$$

In the large-defocus case, Zernike polynomials up to the third order ($N = 9$) together with the primary spherical aberration ($n = 12$) are considered. To reflect the substantial defocus typically observed in experiments, the upper bound for defocus ($n = 4$) is set to 5 rad. The upper bounds V_n are accordingly adjusted as

$$V_n = \begin{cases} 0, & \text{for } n = 0, 1, 2 \\ 5, & \text{for } n = 4 \\ 0.5, & \text{otherwise for } n \le N \\ 0.5, & \text{for } n = 12 \end{cases}. \tag{8}$$

The PICNet model is implemented in PyTorch, with all network weights initialized using a Gaussian distribution with a standard deviation of 0.02. Training is conducted using the Adam optimizer, with hyperparameters set to $\beta_1 = 0.5$ and $\beta_2 = 0.9$. The learning rate is set to 0.00001 for the discriminator and 0.0002 for both the phase and aberration generators. A learning rate decay strategy is employed, reducing the rate by a factor of 0.5 every 50 epochs. The network is trained for 200 epochs with a batch size of 16. Training is performed on a workstation equipped with 32 GB of system memory and an NVIDIA Quadro GV100 GPU, requiring up to 32 h.

4.2 Network Architecture in PICNet

PICNet utilizes the phase generator and aberration generator networks to efficiently predict sample phase distributions and pupil aberrations from captured intensity patterns (see S7 in the Supplementary Material). The phase generator G_θ is a convolutional residual neural network (ResNet) architecture,

composed of an encoder and a decoder. Such network architecture employs downsampling to form a compressed representation and then upsampling to produce the final reconstructed image.[57] The input intensity image is progressively encoded through five sequential modules, each comprising a downsampling block, a residual block, and a dropout layer, then symmetrically decoded through five corresponding modules containing upsampling blocks, residual blocks, and dropout layers. In such an encoder–decoder architecture, skip connections establish direct pathways for low-level features from encoder layers to corresponding decoder layers at the same spatial scale, thereby preserving spatial details and improving pixel-wise reconstruction accuracy. The reconstruction is refined through a final residual block with sigmoid activation, followed by a convolutional layer. The terminal layer of the phase generator directly outputs the estimated phase distribution of the object.

The aberration generator G_ω is implemented using a ResNet-34 architecture, a 34-layer deep residual convolutional neural network originally designed for image recognition tasks.[87] The network is designed to learn a nonlinear mapping from captured aberrated images to corresponding Zernike coefficients that parameterize pupil aberrations. The input aberrated images are initially processed through a convolutional layer followed by max pooling and then passed through four sequential stages of residual blocks with $\{3, 4, 6, 3\}$ layers, enabling progressively deeper hierarchical feature extraction. A global average pooling layer condenses the learned spatial features, which are subsequently transformed by a fully connected layer into a vector of Zernike coefficients that correspond to orthogonal aberration modes. These predicted coefficients are finally utilized to synthesize the pupil aberrations.

In the adversarial learning framework, the discriminator guides the generator toward producing phase distributions that closely approximate the true data distribution in the object domain. Specifically, the phase image is fed into the discriminator that adopts a contracting architecture to extract spatial features. The architecture comprises a sequence of convolutional blocks, each consisting of a convolutional layer, batch normalization, and a LeakyReLU activation function with a negative slope of 0.2. The number of feature channels gradually increases, facilitating multiscale representation of image features. A global feature is subsequently obtained via adaptive average pooling, which is further processed by a final convolutional layer to yield a scalar output that reflects the probability of the input phase belonging to the ground truth distribution. The incorporation of the discriminator in PICNet contributes to improved reconstruction performance.

4.3 Loss Function in PICNet

In the translation cycle from the measurement domain to the object domain and back, the cycle-consistent loss function is computed in the feature domain between the measured and generated intensity,

$$\mathcal{L}_{\mathbf{I}}^{G} = E_{\mathbf{I}\sim p(\mathbf{I})}[\|\Psi\mathbf{I}^{\beta} - \Psi(\mathcal{H}(G_\theta(\mathbf{I}), G_\omega(\mathbf{I})))^{\beta}\|_1], \tag{9}$$

where $\beta = \frac{1}{2}$, E denotes the expectation, Ψ refers to the feature extractor for image structures, and $p(\mathbf{I})$ is the probability distribution of $\mathbf{I}$. By incorporating pattern recognition into the physical framework, the feature-domain loss minimizes the L_1-distance between the features of model predictions and experimental measurements.[88] In our implementation, the first-order edge of images is adopted as feature representations, i.e., $\Psi = [\nabla_u, \nabla_v]^T$ with u and v representing the spatial coordinates. Because certain types of noise and artifacts exhibit distinct statistical properties in the image and feature domains, their impact can be effectively suppressed by separating them in the feature domain of images. The optimization objective shifting from pixel-level errors to feature discrepancies can effectively mitigate the influence of systematic errors.

To ensure cycle consistency from the object domain to the measurement domain and back, a composite loss function that combines pixel-wise and feature-domain terms is applied between the retrieved and ground-truth object:

$$\begin{aligned}\mathcal{L}_{\mathbf{o}}^{G} = {} & E_{\mathbf{o}\sim p(\mathbf{o})}[\|\mathbf{o} - G_\theta(\mathcal{H}(\mathbf{o}, \mathbf{w}))\|_1] \\ & + \lambda E_{\mathbf{o}\sim p(\mathbf{o})}[\|\Psi\mathbf{o} - \Psi G_\theta(\mathcal{H}(\mathbf{o}, \mathbf{w}))\|_1],\end{aligned} \tag{10}$$

where $p(\mathbf{o})$ is the probability distribution of $\mathbf{o}$. λ is equal to $\frac{5}{3}$. A cycle-consistency loss is employed to regularize pupil aberration prediction, encouraging fidelity to the ground truth:

$$\mathcal{L}_{\mathbf{w}}^{G} = E_{\mathbf{w}\sim p(\mathbf{w})}[\|\mathbf{w} - G_\omega(\mathcal{H}(\mathbf{o}, \mathbf{w}))\|_1]. \tag{11}$$

The L_1 loss is chosen to update the network because it offers robustness against data deviations and promotes stable convergence. By contrast, the mean-squared error (MSE, or L_2 loss) applies a quadratic penalty to large deviations, which often produces over-smoothed or blurred outputs. By applying a linear penalty, the L_1 loss preserves sharp edges and structural details. Consequently, adopting the L_1 loss improves optimization stability in cycle-consistent training, allowing reconstructions obtained through domain translations to faithfully recover the original images.

To encourage the predicted outputs to approximate the target solutions, adversarial losses $\mathcal{L}_a^D$ and $\mathcal{L}_a^G$ are utilized to update the network parameters of the discriminator and generators, respectively. The corresponding loss formulations are given as follows:

$$\begin{aligned}\mathcal{L}_a^D = {} & E_{\mathbf{o}\sim p(\mathbf{o})}[\max(0, 1 - D_\phi(\mathbf{o}))] \\ & + E_{\mathbf{I}\sim p(\mathbf{I})}[\max(0, 1 + D_\phi(G_\theta(\mathbf{I})))],\end{aligned} \tag{12a}$$

$$\mathcal{L}_a^G = -E_{\mathbf{I}\sim p(\mathbf{I})}[D_\phi(G_\theta(\mathbf{I}))], \tag{12b}$$

where D_ϕ represents the adversarial discriminator with network parameters ϕ.

Based on the defined loss functions, the PICNet model is trained in an alternating optimization manner, where the generator and discriminator networks are updated iteratively. In each training step, the generator parameters are first held fixed while the discriminator is updated by minimizing the adversarial loss $\mathcal{L}_a^D$ through backpropagation. Then, with the discriminator parameters fixed, the generators are updated by minimizing a composite objective that combines both adversarial and cycle-consistency losses, formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_a^G + \alpha_1\mathcal{L}_{\mathbf{I}}^G + \alpha_2\mathcal{L}_{\mathbf{o}}^G + \alpha_3\mathcal{L}_{\mathbf{w}}^G, \tag{13}$$

where $\alpha_1 = 50$, $\alpha_2 = 60$, and $\alpha_3 = 40$. Therefore, this total objective function is employed to guide the training of both the phase and aberration generators.

4.4 Evaluation Metrics

The fidelity of phase reconstruction is quantitatively assessed using PSNR and SSIM. PSNR measures the pixel-wise reconstruction accuracy by comparing the predicted object $\hat{o}$ with the ground truth o, which is defined as

$$\mathrm{PSNR} = 10\log_{10}\frac{\mathrm{MAX}^2}{\mathrm{MSE}}, \tag{14}$$

where MAX is the maximum pixel value, MSE represents the mean squared error that is $\mathrm{MSE} = \frac{1}{K}\sum_{i=1}^{K}|o_i - \hat{o}_i|^2$, and K denotes the total number of pixels in the phase map. The higher the PSNR value, the better the reconstruction fidelity.

SSIM captures the perceptual similarity between the predicted and ground-truth phase maps by jointly evaluating luminance, contrast, and structural consistency, which is calculated using the following equation:

$$\mathrm{SSIM}(\hat{o}, o) = \frac{(2\mu_{\hat{o}}\mu_o + c_1)(2\sigma_{\hat{o},o} + c_2)}{(\mu_{\hat{o}}^2 + \mu_o^2 + c_1)(\sigma_{\hat{o}}^2 + \sigma_o^2 + c_2)}, \tag{15}$$

where μ, σ, and $\sigma_{\hat{o},o}$ represent the local means, standard deviations, and mutual covariances of $\hat{o}$ and o, respectively, and c_1 and c_2 are stabilization constants. SSIM values range from 0 to 1, with a score of 1 indicating perfect structural agreement between the two images.

The accuracy of the retrieved pupil aberration is quantitatively assessed using the RMSE between the estimated Zernike coefficients $\hat{\mathbf{w}}$ and the ground-truth coefficients $\mathbf{w}$, which is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\|\mathbf{w} - \hat{\mathbf{w}}\|_2^2}. \tag{16}$$

RMSE provides an absolute measure of the overall deviation between the predicted and true aberration parameters. A lower RMSE indicates more accurate aberration estimation.

Disclosures

The authors declare no conflicts of interest.

Code and Data Availability

The codes in this study are available from https://github.com/THUHoloLab/PICNet. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Author Contributions

D. Xu and L. Cao conceived the idea, developed the method, and conducted the simulations and experiments. L. Cao and H. Sun initiated and supervised the project. All authors contributed to the data analysis and the writing of the manuscript.

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